
title: "Dimensional Lumber: Sizes, Conversions & Rules of Thumb" version: "1.0" topic_key: "dimensional_lumber" date: "2025-10-09"

Definitions

BFS (Builders FirstSource) = **BLDR.com**, NYSE: BLDR – a building-products, components and distribution company.

Introduction

Dimensional lumber refers to **softwood boards and timbers that are cut to standardized cross-section dimensions**. These boards are used extensively in residential and light-commercial construction for framing walls, floors, roofs and decks. Unlike rough-sawn lumber, dimensional lumber is planed and surfaced on four sides, so it is ready for immediate use. Because lumber is sawn oversize to account for shrinkage and planing, there is a **difference between the name of a board (its nominal size) and its actual dimensions**. For example, a board sold as a “2×4” started life 2 inches thick and 4 inches wide but after drying and planing it typically measures about **1½ inches by 3½ inches** ¹. Understanding nominal vs. actual dimensions and how to convert between linear footage, board-feet, thousand board-feet (MBF) and thousand square-feet (MSF) helps builders estimate material quantities and cost.

Common Nominal vs Actual Dimensions (2×4, 2×6, etc.)

Nominal and actual sizing

Lumber is described by **nominal sizes** (convenient names such as “2×4” or “2×10”) and by **actual sizes** (the finished dimensions after shrinkage and planing). The Kreg Tool Academy explains that nominal sizes are based on the rough-sawn board before drying, while the **actual size refers to wood that has been planed and trimmed to its final dimensions** ². The difference dates back to early lumber production: boards were cut oversized to allow for shrinkage and were then planed smooth, resulting in final dimensions slightly smaller than nominal ³. Despite the size change, nominal names became industry standard because they allow consistent ordering ⁴. Rough-cut boards may measure closer to their nominal size ⁵.

Standard lengths

Standard dimensional lumber lengths in North America range from **6 ft to 20 ft in two-foot increments**, with longer lengths (up to 24 ft) sometimes available ⁶. By code, the actual length must be at least equal to the nominal length and can be up to 3 inches longer ⁷.

Common dimensional sizes

The table below summarizes the most common 2-by lumber sizes and their actual finished dimensions (Surfaced 4 Sides, S4S) in inches and millimetres ¹. Values are rounded.

Nominal size	Typical actual size (inches)	Typical actual size (mm)
2×2	1.5 × 1.5	38 × 38
2×4	1.5 × 3.5	38 × 89 ⁸
2×6	1.5 × 5.5	38 × 140 ⁸
2×8	1.5 × 7.25	38 × 184 ¹
2×10	1.5 × 9.25	38 × 235 ¹
2×12	1.5 × 11.25	38 × 286 ¹
4×4	3.5 × 3.5	89 × 89 ⁹

These sizes apply to softwood species used for framing. Hardwood lumber uses a “quarters” system (4/4 = 1 inch thick, 8/4 = 2 inch thick) where the widths vary and are usually surfaced by the woodworker ¹⁰.

Why nominal and actual sizes matter

Builders must account for the difference between nominal and actual sizes when designing structures. A 2×4 wall stud is actually 1½ inches thick, so a 10 ft wall framed with 16-inch on-center spacing will have slightly different cavity widths than one might expect from nominal dimensions. When ordering lumber by **board-foot** (volume), the **nominal thickness and width are used**, not the actual size ¹¹. When ordering by **linear foot**, the actual length is used. Misinterpreting the size conventions can lead to shortages or over-ordering.

Conversion Tables & Formulas

Board-foot definition and formula

A **board foot** (bd ft) is a unit of volume equal to a **piece of wood 12 inches long, 12 inches wide and 1 inch thick (144 cubic inches)** ¹². The board-foot content of any board can be determined by multiplying its length, width and thickness (in inches) and dividing by 144 ¹³. Many lumber calculators express the length in feet and the thickness and width in inches; in that case the formula becomes:

$$\text{Board Feet} = \frac{\text{Thickness (in)} \times \text{Width (in)} \times \text{Length (ft)}}{12} \quad (1)$$

This formula appears in multiple industry references ¹⁴ ¹⁵.

When all dimensions are entered in inches, divide by **144** instead of 12:

$$\text{Board Feet} = \frac{\text{Thickness (in)} \times \text{Width (in)} \times \text{Length (in)}}{144}.$$

Linear-feet to board-feet conversion

When purchasing dimensional lumber sold by the piece or by linear foot, builders often need to convert to board feet for pricing. The Western Wood Products Association's conversion table provides the following general formula for converting **linear (linear) feet to board feet**:

$$\text{Board Feet (bf)} = \frac{T \times W \times L}{12} \quad (2)$$

where **T** and **W** are the **nominal thickness and width in inches** and **L** is the length in feet ¹⁶. For example, a 2×6 board (T = 2 in., W = 6 in.) has **1 board foot per linear foot** ($2 \times 6 \div 12 = 1.0 \text{ bf/lf}$). A 2×4 contains 0.667 bf/lf ($2 \times 4 \div 12 = 0.667$). When converting board feet back to linear feet, rearrange equation (2): $L = 12 \text{ bf} / (T \times W)$.

Thousand board-feet (MBF)

Lumber is often traded in **thousand board-feet (MBF)**. Putnam Lumber notes that all MBF calculations are based on the **nominal size**, not the actual size ¹¹. The formula is:

$$\text{MBF} = \frac{T \times W}{12} \times \frac{L}{1000} \times \text{\#pieces} \quad (3)$$

where **T** and **W** are nominal thickness and width (in inches), **L** is length (ft) and **\#pieces** is the quantity ¹¹.

Thousand square-feet (MSF) for panels

Panel products such as plywood and OSB are sold by the **thousand square feet (MSF)**. Thickness is not considered in the MSF calculation. Putnam's conversion guide shows that the number of MSF in a bundle is:

$$\text{MSF} = \frac{\text{Width (ft)} \times \text{Length (ft)}}{1000} \times \text{\#pieces} \quad (4)$$

¹⁷. For example, forty six sheets of ¾-inch 4'×8' plywood (each 4 ft × 8 ft) provide $(4 \times 8) \times 46 \div 1000 = 1.472 \text{ MSF}$ ¹⁷.

Board-feet per linear foot table

Because equation (2) uses nominal dimensions, many contractors memorise board-feet per linear foot for common sizes. The table below (calculated from equation (2)) shows board-feet contained in **one linear foot** of various nominal sizes.

Nominal size	Board feet per linear foot	Nominal size	Board feet per linear foot
1×2	0.1667 bf/lf	2×6	1.0000 bf/lf
1×4	0.3333 bf/lf	2×8	1.3333 bf/lf

Nominal size	Board feet per linear foot	Nominal size	Board feet per linear foot
1×6	0.5000 bf/lf	2×10	1.6667 bf/lf
1×8	0.6667 bf/lf	2×12	2.0000 bf/lf
1×10	0.8333 bf/lf	4×4	1.3333 bf/lf
2×4	0.6667 bf/lf	4×6	2.0000 bf/lf

(Values computed using equation (2) with nominal thickness T and width W .)

Board-Foot & Linear Foot Calculations

This section walks through common calculations using **chain-of-thought (CoT)** reasoning. In practice you would perform the steps mentally or with a calculator; here they are shown explicitly for clarity.

Example 1 – board feet in a 2×4 × 10 ft board

Problem: How many board feet are in one 2×4 board that is 10 ft long?

Step-of-Thought (SoT):

1. Identify the **nominal dimensions**: thickness $T = 2$ inches , width $W = 4$ inches . Remember that board-foot calculations use nominal dimensions ¹¹ .
2. Identify the **length** in feet: $L = 10$ ft .
3. Apply the board-foot formula (equation (2)):

$$\text{Board Feet} = \frac{T \times W \times L}{12} = \frac{2 \times 4 \times 10}{12}.$$

4. Multiply numerator: $2 \times 4 \times 10 = 80$.
5. Divide by 12: $80 \div 12 \approx 6.67$.

Answer: There are ≈ 6.67 board feet in a nominal 2×4 board that is 10 ft long.

Example 2 – converting linear feet to board feet (bundle of 2×6)

Problem: You have 300 linear feet (lf) of nominal 2×6 lumber. How many board feet is that?

CoT reasoning:

1. A 2×6 has nominal dimensions $T = 2$ inches and $W = 6$ inches.
2. Using equation (2), board feet per linear foot is $(2 \times 6) \div 12 = 1$. Therefore each foot of 2×6 contains **1 board foot**.
3. Multiply by the total linear footage: $1 \text{ bf/lf} \times 300 \text{ lf} = 300 \text{ bf}$.

Answer: 300 linear feet of 2×6 lumber contains **300 board feet**.

Example 3 – computing MBF from pieces (2×6 × 16 ft bundle)

Problem: A bundle contains 128 pieces of nominal 2×6 lumber, each 16 ft long. What is the bundle volume in thousand board-feet (MBF)?

CoT reasoning:

1. Determine board-feet per piece using equation (2):

$$\text{bf/piece} = \frac{2 \times 6 \times 16}{12} = \frac{192}{12} = 16 \text{ bf.}$$

2. Multiply by the number of pieces: $16 \text{ bf} \times 128 = 2048 \text{ bf}$.
3. Convert board-feet to MBF by dividing by 1,000: $2048 \text{ bf} \div 1000 = 2.048 \text{ MBF}$.

Answer: The 128-piece bundle contains $\approx 2.048 \text{ MBF}$.

Example 4 – calculating MSF for panel products

Problem: You purchase 46 sheets of ¾-inch plywood measuring 4 ft × 8 ft. How many MSF is this shipment?

CoT reasoning:

1. For MSF calculations, thickness is not considered. Each sheet has area: $4 \text{ ft} \times 8 \text{ ft} = 32 \text{ square feet}$.
2. Multiply by the number of sheets: $32 \text{ sf} \times 46 = 1,472 \text{ square feet}$.
3. Divide by 1,000 to convert to MSF: $1,472 \div 1,000 = 1.472 \text{ MSF}$.

Answer: Forty-six 4×8 sheets total $\approx 1.472 \text{ MSF}$ of panel area.

Rules of Thumb for Use (span, spacing, orientation)

Spanning capacity of dimensional lumber

Safe span depends on species, grade, spacing and loads. Manufacturers publish span tables; the following ranges are taken from **Weyerhaeuser's floor-joist span card** (for #2 and better SPF or southern pine under 40 psf live load). These values assume joists spaced at **16 inches on-center**, a common framing practice

¹⁸.

Nominal joist size	Approximate maximum span range (ft-in)	Source
2×6	$\approx 9'-9'' - 10'-9''$ ¹⁹	Weyerhaeuser floor-joist table
2×8	$\approx 12'-10'' - 14'-2''$ ²⁰	Weyerhaeuser floor-joist table
2×10	$\approx 16'-5'' - 18'-0''$ ²¹	Weyerhaeuser floor-joist table
2×12	$\approx 19'-11'' - 21'-4''$ ²²	Weyerhaeuser floor-joist table

Use these ranges as *rough guidelines* only. Always consult current span tables or a structural engineer, especially for heavy live loads or unusual conditions.

Beam span rule-of-thumb

For beams supporting joists, a simple rule of thumb uses the beam depth (inches) to estimate its span (feet). **Wood It's Real** notes that, for pressure-treated southern pine no. 2 grade, a double-ply beam can span in feet roughly **equal to its depth in inches** ²³. For example, a double 2×12 beam (actual depth ≈ 11¼ inches) can span about **12 ft**, while a double 2×10 can span about **10 ft** ²³. This rule offers a quick sanity check but does not replace engineering calculations.

Wall stud spacing

Traditional wood-framed houses use **2×4 studs spaced 16 inches on-center** ²⁴. Research has shown that **2×6 studs spaced 24 inches on-center** can adequately support exterior walls and are allowed by building codes in many jurisdictions ²⁵. Switching to 2×6 @ 24" OC increases insulation cavity width and reduces thermal bridging, but local codes in hurricane or high-wind zones may still require 16 inch spacing ²⁶.

Crown orientation and grain

Lumber often contains a slight natural **crown** (upward curvature along the length). For beams and joists, it is good practice to **install the crown side up** so that the weight of the structure tends to flatten the member rather than increase the sag ²⁷. The Wood It's Real article explains that cantilevered beams require at least 6 inches of airspace below and should be oriented with the crown up ²⁸. When framing walls, carpenters typically orient studs so that all crowns face the same direction to prevent wavy walls. Proper orientation reduces deflection and facilitates a level finish surface.

Additional framing considerations

- **Advanced framing:** The U.S. Department of Energy's Building America program encourages **24 inch on-center spacing** with 2×6 studs and other techniques (two-stud corners, single top plates) to reduce thermal bridging and save lumber ²⁵.
- **Engineered wood products (EWP):** Engineered products such as **I-joists** offer longer spans and greater dimensional stability than solid lumber. The APA – Engineered Wood Association notes that I-joists are **ideal for long spans** and are manufactured in common depths such as 9½", 11⅞", 14" and 16" ²⁹. Because they are straight and true, builders can avoid crowning issues ³⁰. Considering EWP can reduce material waste and allow open floor plans.

Example Calculations (Worked Examples)

In addition to the board-foot calculations above, here are some other practical examples.

Example 5 – How many lineal feet of 2×8 lumber gives 1 MBF?

1. Determine board-feet per linear foot: a 2×8 has $(2 \times 8)/12 = 1.333 \text{ bf/lf}$.
2. One MBF equals 1,000 board feet. Therefore linear feet required is $1,000 \text{ bf} \div 1.333 \text{ bf/lf} \approx 750 \text{ lf}$.

3. **Answer:** Approximately **750 linear feet** of 2×8 equals 1 MBF.

Example 6 – Converting board feet to cubic meters

Suppose you have **3.321 cubic metres** of surfaced 1×6 lumber (3/4 × 5½ in). Putnam’s metric conversion factor for this size is 0.6167 ³¹.

1. Determine MBF: using equation (3) for 1×6 16’ S4S with 256 pieces, Putnam’s example yields **2.048 MBF** ³².
2. Convert MBF to cubic metres: divide MBF by the metric factor; **2.048 MBF ÷ 0.6167 ≈ 3.321 m³** ³³.
3. **Answer:** 2.048 MBF of 1×6 S4S equates to roughly **3.321 m³**.

Example 7 – Estimating joist span using the depth-rule

A designer is asked to estimate the span for a double 2×10 beam supporting deck joists. Using the rule-of-thumb that a double-ply beam can span in feet equal to its depth in inches ²³:

1. A 2×10 has a nominal depth of 10 inches.
2. Therefore, a double 2×10 beam can span roughly **10 feet**.
3. If the design requires a longer span, consider a double 2×12 (≈ 12 ft span) or engineered wood.
Always verify against manufacturer span tables.

Example 8 – How many board feet are in a 2×12×14’ header?

1. Use equation (2) with nominal dimensions: $(2 \times 12 \times 14)/12 = 28$ board feet.
2. **Answer:** The 2×12×14’ header contains **28 board feet**.

SCAMPER Discussion

The SCAMPER creativity framework encourages thinking about how dimensional lumber might evolve. Here are some prompts:

- **Substitute:** Could sustainably sourced engineered wood replace solid 2×10 joists? I-joists provide longer spans and straighter members ²⁹, potentially reducing the number of intermediate supports.
- **Combine:** What happens when **dimensional lumber is combined with engineered wood products**? Hybrid assemblies (e.g., LVL beams with 2×4 infill) might optimise strength, span and cost.
- **Adapt:** Can we **adapt** standard dimensions to offsite manufacturing? Advanced framing using 2×6 @ 24” OC reduces lumber usage and improves insulation ²⁵.
- **Modify:** How might **new engineered shapes reduce waste**? Continuous cross-laminated timbers or finger-jointed studs could use small offcuts to make long members.
- **Put to another use:** Could short off-cuts be repurposed for blocking, furring or engineered composite products?
- **Eliminate:** Are there components (e.g., jack studs under small headers) that could be reduced with metal hangers or advanced framing techniques?
- **Reverse/Rearrange:** Might reversing the typical orientation (laying 2×4 studs flat rather than upright) create new architectural effects? Always consult structural engineers when experimenting.

Likely User Questions

- **What is the difference between nominal and actual lumber sizes?** Nominal sizes describe rough-sawn dimensions, whereas actual sizes are the smaller finished dimensions after drying and planing ².
- **How many board feet are in a 2×4×10 ft board?** Using equation (2): $(2 \times 4 \times 10)/12 = 6.67$ board feet.
- **How do I convert linear feet to board feet?** Use equation (2); multiply nominal thickness and width (inches), multiply by length (feet), then divide by 12 ¹⁶.
- **How many board feet are in a bundle of lumber?** Multiply board-feet per piece by the number of pieces; divide by 1,000 to convert to MBF ¹¹.
- **What are standard stud spacings?** Traditional framing uses 2×4 studs at 16 inches on-center; codes allow 2×6 studs at 24 inches on-center in many climates ³⁴.
- **How do I calculate MSF for plywood panels?** Multiply the width and length (ft), multiply by the number of sheets and divide by 1,000 ¹⁷.

Related Topics / Crosslinks

- **Engineered Wood Products (EWP)** – LVL, PSL, I-joists and how they compare to dimensional lumber.
- **Glulam & Laminated Beams** – sizes, applications and span tables.
- **Wall Framing Techniques** – stud layouts, advanced framing, and energy-efficient detailing.
- **Deck Construction** – beam and joist sizing, decking materials, fastening methods.
- **Plywood & OSB Panels** – thicknesses, grades and span ratings.
- **Lumber Grades & Species** – understanding species (SPF, SYP, Douglas-fir) and grading rules.

¹ ⁷ ⁸ ⁹ **Lumber Dimensions: Nominal vs Actual - Archtoolbox**

<https://www.archtoolbox.com/lumber-dimensions/>

² ³ ⁴ ⁵ ⁶ ¹⁴ **Ultimate Guide to Lumber Dimensions: Actual vs Nominal Sizes | Kreg Tool**

<https://learn.kregtool.com/learn/understand-lumber-sizing/>

¹⁰ **Lumber Dimensions and Size Measurements**

<https://www.advantagelumber.com/sizes.htm>

¹¹ ¹⁷ ³¹ ³² ³³ **Lumber and Panel Calculations**

<https://www.putnamlumber.com/conversions.html>

¹² ¹³ ¹⁵ **Measuring Standing Trees | Ohioline**

<https://ohioline.osu.edu/factsheet/F-35-02>

¹⁶ **Microsoft Word - Lineal feet to -rsion table.doc**

<https://www.mrslumber.com/wp-content/uploads/2016/06/Lineal-Feet-to-Board-Feet-Conversion.pdf>

¹⁸ ²⁴ ²⁵ ²⁶ ³⁴ **Advanced Framing: Minimum Wall Studs | Building America Solution Center**

<https://basc.pnnl.gov/resource-guides/advanced-framing-minimum-wall-studs>

¹⁹ ²⁰ ²¹ ²² **Lumber Spans Quick Reference Card (U.S.)**

<https://www.weyerhaeuser.com/application/files/3215/4223/2335/LB-4026.pdf>

23 27 28 Spans, Beams, and Joists, Oh My! - Wood. It's Real.

<https://wooditsreal.com/decking/spansbeamsjoists-deck/>

29 30 I-Joist - APA – The Engineered Wood Association

<https://www.apawood.org/i-joist>